BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE – PILANI, HYDERABAD CAMPUS

EEE/INSTR F342: POWER ELECTRONICS, SEM-II, 2026

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Buck Converter and Boost converter

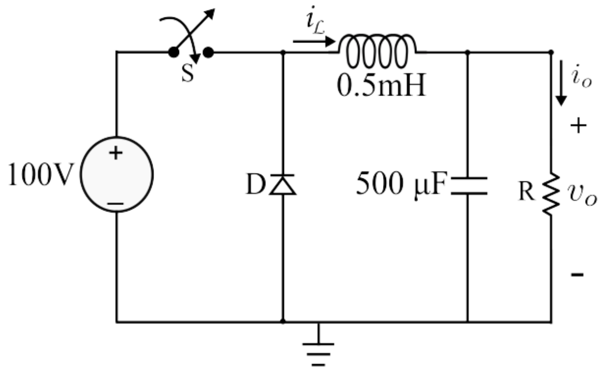
Q1. A buck converter operates with an input voltage of 48 V and a duty ratio of 0.4. The load resistance is 8 Ω. The converter operates in continuous conduction mode. (assume ideal condition)

Find:

a) Output voltage and Output current
b) Average inductor current, Average source current, Average switch current
c) Design the value of inductor, L, such that the inductor current ripple is limited to 20% of average inductor current. Switching frequency = 25kHz.
d) Design the value of capacitor, C such that the output voltage ripple is not exceeding 1% of the output voltage

Q2. A buck converter has the following parameters: Vin=20V, Duty cycle D=0.3, Switching frequency fs=50kHz, Inductor L=40μH, Load resistance R=25Ω. Determine (a)whether the converter operates in continuous or discontinuous conduction mode, (b) The critical inductance for boundary conduction.

Q3. In the dc-dc buck converter shown in the figure, Vin = 100 V and L = 0.5 mH. The duty cycle of the controllable switch (S) is 50 % and its switching frequency is 10 kHz. The average value of the load current is 2 A. Assume that all the components are ideal, and the capacitor and inductor are operating in steady state. Determine (i) the average output voltage. (ii) the respective time durations for which switch S and diode D conduct in each full cycle.



Q4. Draw the circuits and waveforms for (a) the current through the switch, freewheeling diode and the capacitor, (b) voltage across the switch, freewheeling diode and capacitor of a buck converter at discontinuous mode of operation.

Q5. A boost converter has an input voltage of 20V and output voltage of 50 V. The switching frequency is 25 kHz. The inductor L is 200 μH and load resistance is 20 Ω. Assuming ideal Boost converter and continuous current mode, calculate,

1. Duty cycle
2. Average value of input and output currents

3. Inductor peak to peak ripple current
4. Minimum and maximum value of inductor current
5. Filter capacitor ripple voltage if C = 400 μF.